BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 259 · Magnetic, optical and flash storage

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why could a GPU be useful for many independent pixel operations?

Q2. What does ACC hold?

Q3. Define non-volatile.

Q4. Evaluate 11010110_2 AND 00001111_2, then write the resulting byte in hexadecimal.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

magnetic — Bits stored through magnetic states.

optical — Bits encoded/read through optical disc patterns.

flash — Non-volatile electronic storage.

seek latency — Delay in finding data.

backup — Separate recoverable copy of data.

New learning

Identify the storage mechanism — HDDs use magnetised regions on rotating platters; DVD-style media are read optically.

Explain solid-state storage — Flash cells are non-volatile electronic storage without rotating platters or mechanical head seeks.

Evaluate rather than rank universally — SSD generally has low seek latency and resists mechanical shock; HDD may offer lower cost per GB at large capacities.

Worked example

Choose SSD or HDD for a photographer moving large images between sites.

1. Predict — An SSD is attractive for frequent portable access and avoiding moving parts.

2. Explain — An HDD can offer capacity at a lower cost; backups remain necessary regardless.

We Do · Practise with guidance

A school distributes a fixed offline exhibition to computers with disc drives. Would an optical disc work?

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Static media, compatible optical drives.
Read-only distribution is possible.
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Predict

An optical disc can distribute static content without needing a network connection.

Explain

Check disc and drive compatibility and whether updates/reuse are needed.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Give storage technology and one physical feature for HDD, DVD and USB flash.

Question 2

Explain two benefits and one drawback of an SSD rather than an HDD for frequent portable editing.

Question 3

Propose fast working storage and bulk archival storage for a school, explaining backup strategy.

Exit Ticket · Answer before checking

USB flash and HDDs use the same magnetic storage method.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It can execute many similar operations concurrently.

Q2. Intermediate arithmetic/logical result.

Q3. Data remains stored without power.

Q4. $00000110_2 = 06_{16}$.

You Do pages · Independent questions

Q1. [6] HDD magnetic [1] spinning platter [1]; DVD optical [1] laser-read pattern [1]; USB flash semiconductor [1] non-volatile cells [1].

Q2. [6] Faster random access [2], shock resistance due to no moving mechanism [2], typically higher cost per GB [2].

Q3. [6] SSD for regular work/low latency [2]; HDD or suitable archival medium for infrequent bulk data [2]; maintain separate secure backups and check recovery [2].

Exit Ticket · Model answer

False. USB flash uses semiconductor cells.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: